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Phenotypic and genotypic characterization of biofilm forming, antimicrobial resistant, pathogenic *Escherichia coli* isolated from Indian dairy and meat products.

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Abstract:

Escherichia coli are commensal gastrointestinal microflora of humans, but few strains may cause food-borne diseases. Present study aimed to identify antimicrobial resistant (AMR), biofilm-forming *E. coli* from Indian dairy and meat products. A total of 32 *E. coli* isolates were identified and evaluated for biofilm-formation. EMC17, an *E. coli* isolate was established as a powerful biofilm-former that attained maximum biofilm-formation within 96h on glass and stainless-steel surfaces. Presence and expression of virulence-associated genes (adhesins, invasins and polysaccharides) and ability to adhere and invade human liver carcinoma HepG2 cell lines implicates EMC17 to be pathotype belonging to Extra-intestinal Pathogenic *E. coli* (ExPEC). Antibiotic profiling of EMC17 identified it as multi-drug resistant (MDR) strain, possessing extended spectrum β-lactamases (ESBL's) and biofilm phenotype. Early production of quorum sensing molecules (AHLs) alongside EPS production facilitated early onset of biofilm formation by EMC17. Furthermore, the biofilm-forming genes of EMC17 were significantly upregulated 3-27 folds in biofilm-state. This study showed prevalence of MDR, biofilm forming, pathogenic *E. coli* in Indian dairy and meat products that potentially serve as reservoirs for transmission of antimicrobial-resistant (AMR) genes of bacteria from food to humans and pose serious food safety threat.

Keywords: *E.coli*, Biofilms, Antimicrobial resistance (AMR), Food safety

1. INTRODUCTION:

The incidences of food-borne illnesses and food intoxications have been constantly rising due to changing lifestyle, industrial production and global trade of food products (Jones et al., 2017). Pathogen laden foods pose a serious threat and challenge to: the food manufacturers due to contamination and food spoilage; the consumers who suffer high toll of morbidity and mortality; as well as to the government which is responsible for the public health and safety. Lack of prevalence data on food-borne diseases in developing countries like India, pose serious challenge towards disease surveillance and risk analysis in foods.

Escherichia coli are commensal gastrointestinal microflora of humans providing support in digestion and biosynthesis of some vitamins (Sarowska, J. et al, 2019). While most strains are harmless, few *E. coli* strains are pathogenic causing intestinal or extra-intestinal infections of varying severity, primarily transmitted via consumption of contaminated food (Liu et al., 2018; Nordstrom et al., 2013). They may be difficult to detect and treat due to their short doubling time, adherence ability to surfaces and biofilm formation. Presently, limited information is available on occurrence of *E. coli* in the Indian food industry. However, it is reported to be the root-cause for more than 63,000 deaths out of 4,20,000 of total mortality caused by food-borne diseases (WHO, 2018). Presence of *E. coli* in food makes it unfit for human consumption leading to product rejections, economic losses and threatening food security.

A biofilm is defined as an organized collection of surface attached microbial communities of cells that are embedded into a self-produced exopolymeric matrix mainly composed of proteins, polysaccharides and sometimes DNA (Hall-Stoodley et al., 2004). The formation of biofilm is a result of different stress condition(s) where biofilms acts as a defense mechanism enhancing the survival rate of microorganism. They play an important role in microbial pathogenesis and persistence as well as serve as grounds for genetic exchanges. It acts as a shield protecting the microbial community from action of various antimicrobial agents such as antibiotics, preservatives, chemical sanitizers, thermal treatment etc. that are traditionally used in food industry, thus making them robust and hard to eradicate (Monte et al., 2014). The biofilm formation for many bacterial species including *E. coli* occurs as early as two hours and they can survive up to ten years in food industries despite the regular cleaning and sanitation treatment (Corcoran et al., 2014).

Though most sectors of food industries are vulnerable to adverse effect of biofilms, animal food sector encompassing the dairy and meat, takes the hardest hit. The heavy usage of antibiotics in animal industry for disease prevention, has resulted in development of multi-drug resistant (MDR) strains of different pathogenic bacteria which are hard to eradicate and difficult to alleviate clinically (Casburn-Jones and Farthing, 2004). The spread and transmission of MDR bacteria bearing antimicrobial resistance (AMR) genes is a major challenge in treatment and disease control.

The present study aims at gaining valuable insights into the prevalence of food-borne pathogenic *E.coli* in Indian dairy and meat products and in particular the occurrence of biofilm formation and multi-drug resistant strains. Furthermore, detailed research has been conducted to characterize one of the selected biofilm forming, multi-drug resistant, potentially pathogenic strain of *E. coli,* which may pose as a threat of transmitting disease and AMR genes to humans via food and consequent food safety concerns.

2. Materials and Methods

2.1 Isolation and biochemical identification of E. coli

E. coli strains from 30 samples each of dairy products (Indian soft cheese- Paneer, Indian

Sweet- Rasmalai and Raw Milk) and poultry meat (raw chicken) were isolated and identified according to ISO 8261 and ISO 6887-2 respectively. Isolations were made during period of January to June (2014) from samples collected from local markets in Delhi and Haryana, India. The isolates were further confirmed biochemically using HiMedia's Hi*E.coli* identification kit–KB010.

2.2 Molecular Identification and characterization

Biochemically suspected strains were validated by PCR using *rrsH* primers specific to 16srDNA of *E. coli* (Simmons et al., 2014). The *E.coli* isolate that tested positive by PCR was further sequenced (Eurofins, Bangalore) by 16srDNA sequencing using universal primers (Table S1) as described by Marchesi et al. (1998). The sequence was aligned for similarity using BLASTn software on NCBI servers to confirm the species. The phylogenetic tree was developed using bioinformatics tool Molecular Evolutionary Genetic Analysis (MEGA-X), using the neighbor-joining method with maximum composite substitution model. The topological accuracy of the tree was evaluated using the bootstrap test (500 replicates) with evolutionary distance computed using maximum composite likelihood method. Additionally, the molecular identification of antibiotic resistance (blaTEM, blaSHV, blaCMY2, CTXM1) and virulence (*stx1, stx2, hly, eae, fimA, csgA, malA, csgG, tabA*) genes for STEC, EHEC and extraintestinal pathogenic E. coli (ExPEC) was performed using PCR (Sarowska, J. et al, 2019; Simmons et al., 2016; Štaudová et al., 2015; Uhlich et al., 2009; Meng et al., 1998). All primers used in this study are listed in Supplementary Table 1.

2.3 Antibiotic susceptibility testing

Antibiotic susceptibility of isolate towards 18 different clinically relevant antibiotics was evaluated using disc diffusion method on nutrient agar plates. The inhibition zone was and strains were classified as susceptible, intermediate and resistant in accordance to the criteria described by Clinical and Laboratory Standard Institute (CLSI, 2016).

2.4 Evaluation of biofilm formation

2.4.1 Crystal Violet (CV) Assay

Biofilm formation was evaluated using CV assay as described by Stepanovic, Vukovic, Dakic, Savic, & Svabic-Vlahovic, (2000).. The bacteria were classified into 4 groups i.e. strong ($4\times OD_c<OD$), moderate ($2\times OD_c<OD\leq 4\times OD_c$), weak ($OD_c<OD\leq 2\times OD_c$) and non-adherent ($OD\leq OD_c$) on the basis of their biofilm-forming potential. The mean OD_{540} obtained from three replicates was measured and value of negative control was considered as cut-off OD (ODc). The rate of change in OD_{540}, corresponding to biofilm formation rate (BFR), was calculated for different time points using Microsoft excel.

2.4.2 Epifluorescent microscopy

Clean, sterile glass slide and SS316 stainless steel coupons were used to visualize biofilm formation as per the protocol described by Little et al., 2015. Briefly, vertically submerged slides/coupons were placed in 50 ml tubes containing 25 ml of 0.1 OD_{600} culture and incubated under static conditions at 37°C for different time points. This resulted in formation of biofilms at the air liquid interface. After each incubation, slides were removed from growth medium, washed twice in sterile PBS and fixed with 2.5% glutaraldehyde for 30 minutes followed by staining with Fluorescene Di-Acetate (FDA) for 6 minutes in dark (Taneja et al., 2010). Thereafter, slides were rinsed with PBS, dried and observed under the

FITC filter of Nikon Eclipse Ci, epifluorescent microscope (Japan).

2.4.3 Scanning Electron Microscopy

The glutaraldehyde fixed slides (as in section 2.4.2) were taken, dehydrated by sequential ethanol treatment, vacuum dried and cut into 2mm specimens, followed by coating with gold particle in sputter coater (JEOL JFM 1100). The images were captured using digital scanning microscope JSM 6100 (JOEL) with accelerating voltage of 0.3 to 15 kV.

2.5 Quantification of exopolysaccharide (EPS) produced

EPS was extracted from biofilm-state of selected strain by procedure as described by Jung, Choi, & Lee, (2013). The extracted EPS was quantified using phenol sulfuric method as described by Dubois et al., (1956) wherein total EPS was estimated by using standard curve for different dextrose concentrations (in µg/ml). The experiment was repeated three times and resultant value represents the mean of these values.

2.9 Estimation of Quorum-Sensing molecule.

The production of quorum sensing molecule, N-acyl homoserine lactones (AHL) at different time points was evaluated for selected strains according to the protocol described by Thakur et al., (2016). Formation of a dark brown color at end point indicated for presence of lactone compounds in the samples.

2.10 Gene expression studies

Microbial culture (0.1 OD_{600}) was statically incubated in petridishes with three replicates at 37°C for 24h. After incubation, the planktonic cells were collected separately, and the adhered cells were washed with 1X PBS (to remove weakly attached cells), scrapped-off using cell scrapper and collected. The planktonic and adhered cells were centrifuged at 8000 rpm for 5 minutes and the cell pellets were used for isolation of total RNA using RNeasy mini kit (Qiagen Technologies, Germany). Two-hundred nanograms RNA was used to synthesize cDNA using high-capacity cDNA reverse transcription kit (Applied Biosystems) using manufacturers protocol. The RNA and cDNA was quantified spectrophotometrically using NanoDrop-One (Thermo Scientific). qRT-PCRs were performed for various gene targets using the Power UP^{TM} SYBR Green master mix (Applied Biosystem) in CFX96 Touch Real-Time PCR (Bio-Rad). Amplification specificity was confirmed by melt down curve analysis. The gene expression levels were analyzed using livak method after normalization using housekeeping gene *rrsH* and expressed as mean of fold-expression change in biofilms as compared to planktonic-state.

2.11 ***Adhesion-Invasion Assay***

The virulence of the EMC17 strain was determined by performing invasion assays on human liver carcinoma cell lines HepG2. The non-pathogenic laboratory strain *E. coli* DH5α and uropathogenic *E. coli* MTCC729 were taken as reference controls. Standard protocols as described previously (Reddy and Austin, 2017) were followed. The human liver carcinoma cell line (HepG2) was seeded in wells of a 12-well plate at an initial cell density of 10^5 cells/well (growth area 3.9cm^2) in high glucose DMEM supplemented with 10% FBS without any antibiotics and were incubated at 5% CO_2, 37°C for 24 h. Mid-log phase bacteria (*E. coli* EMC17, *E. coli* DH5α and *E. coli* MTCC 729) were added to each well at a Multiplicity of Infection (MOI) ~1:100. The infected monolayers were incubated (37°C at 5%

CO_2) for 2h for adhesion followed by washing and further incubation in Gentamycin (100 µg/mL) containing medium for another 2h for invasion assay. Following the incubation the monolayers were washed three times with sterile 1X PBS and the cells were lysed using 0.1% Triton X-100. The released intracellular bacteria were enumerated by serial dilution and plating on LB agar plate and incubated at 37°C. Each assay was performed in three replicates. Results are presented as Log10 CFU/ml (mean ± SD).

2.12 Statistical Analysis

All the results are expressed for biofilm formation, EPS production, Quorum Sensing and gene expression are represented as mean of three replicates along with standard deviation. The statistical significance of the results (wherever required) has been evaluated using one-way ANOVA with *Dunnett's post hoc* test using GraphPad Prism version 7 software. P <0.05 was considered to be statistically significant

3. Result and Discussion

3.1 Isolation and Identification of isolate using biochemical methods

A total of 32 strains of *E. coli* across 60 dairy and meat samples were isolated and biochemically identified (Supplementary Table S2). Eighteen strains from poultry meat (60% prevalence) and others from dairy products such as raw milk (6 isolates, 60% prevalence), Indian soft cheese-paneer (4 isolates, 40% prevalence) and Indian dairy sweet- rasmalai (4 isolates, 40% prevalence) were isolated. This information on prevalence of *E. coli* in Indian dairy and meat products is significant from food safety and risk analysis perspective.

3.2 Biofilm formation of isolated strains

Biofilm formation is one of the most important virulence factors that protect microbes from antimicrobial drugs and treatments (Olsen, 2015). The ability of spoilage and pathogenic bacteria to adhere onto food surfaces and form biofilms serve as a persistent source of food contamination, threatens food safety and causes huge losses to the food industry (Tezel and Şanlıbaba, 2018). Therefore combating microbial biofilms is a major challenge to food industry (Hall and Mah, 2017). Taking this into consideration, all 32 isolates of *E. coli* were evaluated for their potency to form biofilm on a polystyrene surface. It is evident from figure 1(a), that out of 32 isolates obtained, 4 were strongest (S) (EMC17, EMC14, EMC23 and EMC10), 11 were moderate (M), 15 weak (W) biofilm-formers and 2 non-adherent. EMC17, a powerful biofilm-forming isolate from poultry meat, was selected and studied further for its biofilm-forming kinetics at different time points (Figure 1b). The study revealed that EMC17 possesses a high capacity of adherence to plastic surfaces (polystyrene) at early time point (2h) and establish biofilms within 24h. The BFR as calculated from its direct relation with OD_{540} showed that EMC17 biofilm grew with maximum rate of 0.015 h^{-1}. Though there is continuous biofilm formation reaching maximum at 96h yet the rate of biofilm formation was observed to drastically reduce after 24h, suggesting their transition into quasi-dormant state. Numerous studies have shown that *E. coli* possess an ability to attach and form biofilm on varied surfaces including food contact surfaces such as stainless steel, PVC, polystyrene, polypropylene, glass etc. and resist chemical, pressure and heat treatments used in food industries (Galié et al., 2018; Carter et al., 2016). The high degree of persistence and resistance facilitates proliferation of biofilm-forming *E. coli* into food product thereby making them a leading cause of food-borne outbreaks globally (Galié et al., 2018).

3.3 Molecular Confirmation of Isolates

To establish phylogenetic origin of EMC17 and its characterization, PCR for 16s rDNA (rrsH) gene along with sequencing and homology alignment was performed. The presence of rrsH amplicon (200bp) established it to be *E. coli* (Figure S2a). Furthermore, 16srDNA sequencing of a 1300 bp amplicon (obtained using universal primers), followed by homology match of sequence, performed by nBLAST (NCBI), showed 97% similarity index of EMC17 to *E. fergusonii*, *E. alberti* and *E. coli.* The sequence has been submitted to NCBI GenBank (Accession no. MT772280). This sequence was further used to develop phylogenetic tree using MEGA-X software. The phylogenetic analysis indicated EMC17 was closely linked to *E. alberti* (Figure S2b). Maheux et al., (2011) have previously reported that environmental *E. coli* strains (comprising *E. albertii* and *E. fergusonii,*) are often indistinguishable phenotypically and phylogenetically from typical *E. coli* using 16srRNA sequencing. Therefore, to validate the species of EMC17, we performed PCR for *uidA* gene, that encodes for a highly conserved sequence in *E. coli* and has been proposed as a concrete method to discriminate non-*E. coli* strains from *E. coli* (Maheux et al., 2011). EMC17 was confirmed to be *E. coli* by using *uidA* gene specific PCR yielding specific amplicon of 1487bp (Figure S2c).

3.4 Antibiotic Susceptibility

The multi-drug resistance in pathogens has been recognized as a severe public health hazard globally (Interagency Coordination Group on Antimicrobial Resistance (IACG), 2019). A possible mechanism for being antibiotic resistant is the ability of bacterial subpopulation to transform into a biofilm-state with unique protective phenotype similar to spore-formation (Ito et al., 2009). Therefore, antibiotic sensitivity profile of EMC17 was carried for 18 antibiotics of clinical relevance (Table 1). EMC17 was found to be completely resistant against Clindamycin, Erythromycin, Amoxicillin, Ampicillin, Oxacillin and Cephalothin, majorly belonging to lincosamides, macrolides and β-lactam family of antibiotics. Since, antibiotic resistance is reportedly enhanced by biofilm formation, we believe resistance exhibited by EMC17 is a phenotypic shield created by EPS produced by biofilm forming *E. coli* to prevent antibiotics from reaching their targeted cell. Similar study by Karigoudar et al., (2019) and Sudheendra and Basavaraj, (2018) showed increased resistance of *E. coli* to Co-trimoxazole, Gentamicin, and Imipenem compared to non-biofilm producers.

Additionally, EMC17 was assessed for presence of AMR genes responsible for production of ESBLs, a group of enzymes linked to resistance towards β-lactam class of antibiotics. Simmons et al., (2016) reported that presence of blaTEM, blaSHV, blaCMY2, CTXM1 loci is linked to β-lactam resistance in food-borne *E. coli*. The PCR for blaTEM, blaSHV and CTXM1 did not yield any amplifications indicating for their absence in EMC17. However, specific bands were obtained with primers for blaCMY2 gene (supplementary Figure S1), indicating for genetic presence of this β-lactamase enzyme. This corroborates well with the results of disk diffusion assays (Table 1). Touzain et al., (2018) reported that blaCMY2 gene (plasmid-borne entity) enhances the probability of transmission of these AMR loci to other related pathogenic microbes within food environment or other milieu, thus transforming other microorganisms into antibiotic resistant phenotype and posing as a serious food safety threat.

Therefore, EMC17 possesses both genotypic and phenotypic resistance towards broad class of antibiotics that may get transmitted to other bacteria present in food milieu. The bacterial biofilms are indeed known to serve as favorable locations to facilitate genetic exchanges amongst inter- and intra-species microorganisms, evolving them into a population of multidrug resistant superbugs (Molin and Tolker-Nielsen, 2003).

3.5 Detection of virulence associated genes in E. coli EMC17

Many of *E. coli* strains are reported to form biofilms and commonly cause infections of the intestinal tract (gastroenteritis) and extra-intestinal host sites causing diseases such as urinary tract infections (UTIs), sepsis and neonatal meningitis (Sarowska et al., 2019). EMC17 was evaluated for presence of major virulence genes associated with pathogenic *E. coli* belonging to shiga-toxin producing *E. coli* (STEC), enterohemorrhagic *E. coli* (EHEC) and extra-intestinal pathogenic *E. coli* (ExPEC). The genetic loci tested by PCR included: *stx1, stx2* (encoding shiga-like toxin of STEC) ~~and~~ *eae, hly* (encoding intimin and α-hemolysin of EHEC; Meng et al., 1998) and *csgG* (encoding invasins), *malA* (encoding polysaccharides) and *tabA* (encoding adhesins) of ExPEC (Sarowska et al., 2019). EMC17 did not possess *stx1, stx2,eae, hly* genes (data not shown), however, all the tested markers for ExPEC (*csgG, malA, tabA, csgA, fimA*) were found to be present in EMC17 strain (data not shown), hence pointing towards it being an ExPEC strain of *E. coli*. ExPEC are a pathotype of *E. coli* commonly occurring in a wide range of hosts. Many ExPEC strains are of animal origin (Bergeron et al., 2012; Johnson et al., 2017), commonly occurring as contaminants in pork, beef, poultry meat and ready to eat foods products and are responsible for increased morbidity and mortality in humans and animals (Li et al., 2018; Xu et al., 2019). In humans, they majorly cause infections of the extra-intestinal organs causing urinary tract infections (subpathotype UPEC), neonatal bacterial meningitis (subpathotype NMEC), septicemia etc. In birds especially poultry a subpathotype of ExPEC called APEC (Avian pathogenic *E. coli*) cause colibacillosis, putting a significant economic burden on the poultry industry. Studies have confirmed that APEC isolates from broiler chicken feces are virulent and get transmitted causing urinary tract infection in mouse models (Stromberg et al., 2017, Sarowska et. al., 2019). The ability of most APEC strains to form biofilms and their persistence in the poultry production environment (due to biofilm development) on plastic surfaces such as water feeding systems may be the underlying cause of disease transmission in poultry (**Lindsay et al.,1996; Amaral, 2004**). Many studies have found overlying characteristics and gene sequences between APEC and human ExPEC, leading to the hypothesis of a zoonotic potential of poultry strains **(Borges et al., 2019)**. Further the ExPEC strains have a complex phylogenetic structure with wide range of virulence factors which includes toxins, adhesins, iron acquisition factors lipopolysaccharides, polysaccharides capsule and invasins usually encoded on pathogenicity islands (Sarowska, J. et al, 2019). CsgA is essential for invasion of cultured HEp-2 cells (Uhlich et al., 2009), whilst FimA encodes adherence factor for Type 1 fimbriae (Štaudová et al., 2015) important for microbial attachment to host surfaces. The problematic increase in the cases of UTIs by Uropathogenic *E. coli* (UPEC) and increase in antibiotic resistance have resulted in huge difficulties in treatment and cure of ExPEC-mediated infections. About 80% of UTIs recorded in the United States are due to ExPEC, and they account for over 4-5 billion dollars costs per annum over healthcare. Moreover, the capacity to form biofilms by some ExPEC *E. coli* strains acts as additional virulence factor and are difficult to eliminate (Safadi et al., 2012).

3.6 Invasion of EMC17 strain

Attachment followed by invasion of bacteria is a vital feature indicating towards virulence of pathogenic bacteria. The EMC17 showed significantly higher invasion of HepG2 cells as compared to the non-pathogenic laboratory strain *E. coli* DH5α (Fig. 5). Further the invasion capability of EMC17 matched that of the reference uropathogenic strain of *E.coli* MTCC 729 showing no statistically significant difference and achieving high levels of intracellular bacterial load. This indicates that *E. coli* EMC17 is a highly invasive potential pathogenic strain.

In view of the presence of virulence genes for ExPEC in EMC17 meat isolate, it's increased ability to adhere and invade HepG2 cells, resistance to multiple antibiotics and biofilm forming ability; altogether makes it a potential candidate of concern to food safety. Detailed investigations are required to evaluate its pathogenesis, transmission and infectivity in humans and animals. Nonetheless, the prevalence of this strain in meat products in the Indian market is a serious concern and a major challenge to food safety and public health, both at production level and in course of their processing and distribution.

3.6 Microscopic study of Biofilms

Microscopic studies (Fluorescent and SEM) are a powerful tool for determining fine structure of bacterial system and biofilms. In the present study we analyzed biofilm evolution as a function of time using fluorescent microscopy and Scanning Electron Microscopy study (Figure 4a & 4b). The microscopic imaging (SEM and Fluorescent Microscopy) of lawn of biofilm on glass slide depicted numerous rod shaped bacterial cells embedded within EPS matrix. This pattern of biofilm formation was observed at various time intervals of 24, 48, 72 and 96h of incubation. It is clearly evident from the microscopy images that biofilm structures become thick and denser over a period of time. This is in accordance with previous study conducted by Lüdecke et al., (2014) indicating that *E. coli* biofilm grows denser over a period of time on titanium oxide and glass surfaces respectively.

3.7 Quorum Sensing and EPS production

QS helps bacteria to coordinate with neighboring bacterial cells to accomplish cooperative activities such as bioluminescence, sporulation, competence, antibiotic production, biofilm development and virulence factor secretion etc. (Rutherford and Bassler, 2012). Small chemical molecules like Acyl Homoserine Lactone (AHLs) play a vital role in cell communications and quorum sensing in gram-negative bacteria, which are necessary for initial biofilm formation (Bassler, 1999). The EMC17 was evaluated for ability to produce AHL molecules at early time points of biofilm formation. Figure 5(a) shows the presence of QS molecule as a function of time. It is clearly evident from the figure that EMC17 showed a temporal increase in QS activity (production of AHLs) upto 8h of incubation when compared to non-biofilm forming strains, EMC 8 and ATCC25922, which showed negligible presence of AHLs molecules during the incubation period.

Furthermore, EMC17 was tested for its ability to produce EPS (a complex mixture of biopolymers formong the supporting three dimensional structure of biofilms). Figure5(b) shows increase in EPS level from 25µg/mL to 45µg/mL at 24h and 96h of incubation period respectively. The EPS production in case of non-biofilm forming EMC8 and ATCC 25922 strains, remained practically constant and negligible. This is in agreement with microscopy study (Figure 4b) and assimilates with observed increase in biofilm formation of EMC17 over 24-96h (Figure 1b).

3.8 Gene Expression Analysis

Biofilm formation is a multi-staged process involving regulations of several genes that play an important role to accomplish a resistant phenotype which is physiologically more adaptable. We studied the relative expression of six selected genes (*fimA, csgA, luxS, malA, bssS and bssR*) reported to be involved in biofilm formation of *E. coli* (Schembri et al., 2003). All tested genes were upregulated in the biofilm-state (Figure 6).

The LuxS enzyme, encoded by *luxS* gene in *E. coli*, plays a significant role in synthesis of quorum sensing molecule i.e. auto inducer-2 (AI-2) which plays a pivotal role in initiating cell-cell communication at early stages of biofilm formation (Zhu et al., 2006). We observed a 27-fold increase in expression of *luxS* gene in EMC17 under biofilm conditions. Wang et al., (2012) reported that an overexperesssion of LuxS synthesized more active AI-2 that would thereby lead to higher biofilm formation by bacterial strains. The increased expression of *luxS* is warranted by our findings of increased production of QS and AHL molecules with an intense biofilm formation which would only be possible to achieve via enhanced expression levels of biofilm-forming genes like *luxS*.

The genes *fimA* and *csgA* encodes for major subunits of type-1 fimbriae and curli protein respectively. Both genes are known to be responsible for initial and irreversible attachment to surfaces during primary phases of biofilm formation. These genes were highly upregulated in EMC17, with *fimA* and *csgA* showing 23 and 15-fold increase respectively. Beloin et al., 2004, reported upregulation of *fimA* by 3.23 folds in *E. coli* K-12 biofilm. The differences in fold upregulation, may be attributed to differences in sources of these bacteria as well as culture conditions that influence the physiological state of bacteria. EMC17 is a food isolate which was exposed to different processing and environmental conditions while *E. coli* K-12 is a laboratory strain, that thereby influences their respective expression profiles. *malA* gene encodes for 4-alpha-glucanotransferase, which is a signaling molecule controlling maltose uptake and utilization. This gene along with other genes present on the same operon are upregulated during biofilm formation (Creti et al., 2006). Expression of *malA* was upregulated by 2.5 fold in EMC17 biofilms. Genes *bssS* (*yceP*) and *bssR* (*yliH*) which play a significant role in regulation of biofilm by cell signaling were also upregulated by 2 and 7 folds respectively. The *bssS* (*yceP*) and *bssR* (*yliH*) genes are known for their role in regulation of signal secretion (Domka et al., 2006). These gene loci are well studied and reported to be induced during biofilm formation by fold increase in expression by 3.06 and

2.97 times respectively in *E. coli* K12 strain (Beloin et al., 2004). Our results on expression of major genes involved in *E. coli* biofilm formation correlate well with other published data, thereby establishing EMC17 as a strong biofilm forming strain of *E. coli* isolated from meat product.

The data presented in this study valuably contributes to the knowledge on prevalance of multi-drug resistant, biofilm forming *E. coli* in Indian dairy and meat sector. The isolated strain EMC17 expresses genes for biofilm formation and bears antibiotic resistance genes that may get transmitted to other potential pathogens in food or humans and pose a serious food safety threat and health hazard. Further studies on growth and survival of this isolate under varied processing conditions is important to understand and estimate the risk of disease transmission and potential threat to consumers.

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Conflict of Interest

All authors declare no conflict of interest.

References

Amaral, L.D., 2004. Drinking water as a risk factor to poultry health. Rev. Bras. Ciên. Avíc.

6, 191–199. doi: 10.1590/S1516-635X2004000400001

Bassler, B.L., 1999. How bacteria talk to each other: Regulation of gene expression by quorum sensing. Curr. Opin. Microbiol. 2, 582–587. https://doi.org/10.1016/S1369-5274(99)00025-9

Beloin, C., Valle, J., Latour-lambert, P., Faure, P., Kzreminski, M., Balestrino, D., Haagensen, J.A.J., Molin, S., Prensier, G., Arbeille, B., Ghigo, J., 2004. Global impact of mature biofilm lifestyle on *Escherichia coli* K-12 gene expression. Mol. Microbiol. 51, 659–674. https://doi.org/10.1046/j.1365-2958.2003.03865.x

Bergeron, C.R., Prussing, C., Boerlin, P., Daignault, D., Dutil, L., Reid-Smith, R.J., Zhanel, G.G., Manges, A.R., 2012. Chicken as reservoir for extraintestinal pathogenic *Escherichia coli* in Humans, Canada. Emerg. Infect. Dis. 18, 415–421. https://doi.org/10.3201/eid1803.111099

Borges, C.A., Tarlton, N.J., Riley, L.W., 2019. *Escherichia coli* from Commercial Broiler and Backyard Chickens Share Sequence Types, Antimicrobial Resistance Profiles, and Resistance Genes with Human Extraintestinal Pathogenic *Escherichia coli*. Foodborne Pathog Dis. 16, 813-822. doi:10.1089/fpd.2019.2680

Carter, M.Q., Louie, J.W., Feng, D., Zhong, W., Brandl, M.T., 2016. Curli fimbriae are conditionally required in *Escherichia coli* O157: H7 for initial attachment and biofilm formation. Food Microbiol. 57, 81–89. https://doi.org/10.1016/j.fm.2016.01.006

Casburn-Jones, A.C., Farthing, M.J.G., 2004. Management of infectious diarrhoea. Gut 53, 296–305. https://doi.org/10.1136/gut.2003.022103

CLSI, 2016. Performance standards for antimicrobial susceptibility testing, Clinical Microbiology Newsletter. https://doi.org/10.1016/s0196-4399(01)88009-0

Corcoran, M., Morris, D., De Lappe, N., O'Connor, J., Lalor, P., Dockery, P., Cormican, M., 2014. Commonly Used Disinfectants Fail To Eradicate *Salmonella enterica* Biofilms from Food Contact Surface Materials. Appl. Environ. Microbiol. 80, 1507–1514. https://doi.org/10.1128/aem.03109-13

Creti, R., Koch, S., Fabretti, F., Baldassarri, L., Huebner, J., 2006. Enterococcal colonization of the gastro-intestinal tract: Role of biofilm and environmental oligosaccharides. BMC Microbiol. 6, 1–8. https://doi.org/10.1186/1471-2180-6-60

Domka, J., Lee, J., Wood, T.K., 2006. YliH (BssR) and YceP (BssS) Regulate *Escherichia coli K-12* Biofilm Formation by Influencing Cell Signaling †. Appl. Environ. Microbiol. 72, 2449–2459. https://doi.org/10.1128/AEM.72.4.2449

Dubois, M., Gilles, K.A., Hamilton, J.K., Rebers, P.A., Smith, F., 1956. Colorimetric Method for Determination of Sugars and Related Substances. Anal. Chem. 28, 350–356. https://doi.org/10.1021/ac60111a017

Galié, S., García-Gutiérrez, C., Miguélez, E.M., Villar, C.J., Lombó, F., 2018. Biofilms in the food industry: Health aspects and control methods. Front. Microbiol. 9, 1–18. https://doi.org/10.3389/fmicb.2018.00898

Hall-Stoodley, L., Costerton, J.W., Stoodley, P., 2004. Bacterial biofilms: From the natural environment to infectious diseases. Nat. Rev. Microbiol. 2, 95–108. https://doi.org/10.1038/nrmicro821

Hall, C.W., Mah, T.F., 2017. Molecular mechanisms of biofilm-based antibiotic resistance and tolerance in pathogenic bacteria. FEMS Microbiol. Rev. 41, 276–301. https://doi.org/10.1093/femsre/fux010

Interagency Coordination Group on Antimicrobial Resistance (IACG), 2019. NO TIME TO WAIT : SECURING THE FUTURE FROM DRUG-RESISTANT INFECTIONS.

Ito, A., Taniuchi, A., May, T., Kawata, K., Okabe, S., 2009. Increased Antibiotic Resistance of *Escherichia coli* in Mature Biofilms . Appl. Environ. Microbiol. 75, 4093–4100. https://doi.org/10.1128/AEM.02949-08

Johnson, J.R., Porter, S.B., Johnston, B., Thuras, P., 2017. Extraintestinal Pathogenic and Antimicrobial Resistant *Escherichia coli* from Retail Chicken Breasts in the United States in 2013. Appl. Enviromental Microbiol. 83, 1–9. https://doi.org/10.1128/AEM.02956-16

Jones, A.K., Cross, P., Burton, M., Millman, C., O'Brien, S.J.O., Rigby, D., 2017. Estimating the prevalence of food risk increasing behaviours in UK kitchens. PLoS One 1–17. https://doi.org/10.5061/dryad.5mh53.Data

Jung, J.H., Choi, N.Y., Lee, S.Y., 2013. Biofilm formation and exopolysaccharide (EPS) production by *Cronobacter sakazakii* depending on environmental conditions. Food Microbiol. 34, 70–80. https://doi.org/10.1016/j.fm.2012.11.008

Karigoudar, R.M., Karigoudar, M.H., Wavare, S.M., 2019. Detection of biofilm among uropathogenic *Escherichia coli* and its correlation with antibiotic resistance pattern. J. Lab. Physicians 11, 17–22. https://doi.org/10.4103/JLP.JLP

Li, B., Huang, Q., Cui, A., et al., 2018. Overexpression of Outer Membrane Protein X (OmpX) Compensates for the Effect of TolC Inactivation on Biofilm Formation and Curli Production in Extraintestinal Pathogenic *Escherichia coli* (ExPEC). Front Cell Infect Microbiol. 8, 208. doi:10.3389/fcimb.2018.00208

Lindsay, D., Geornaras, I., von Holy, A., 1996. Biofilms associated with poultry processing equipment. *Microbios*.86, 105-116.

Liu, C.M., Stegger, M., Aziz, M., Johnson, T.J., Waits, K., Nordstrom, L., Gauld, L., Weaver, B., Rolland, D., Statham, S., Horwinski, J., Sariya, S., Davis, G.S., Sokurenko, E., Keim, P., Johnson, J.R., Price, L.B., 2018. *Escherichia coli* ST131-H22 as a foodborne uropathogen. MBio 9, 1–11. https://doi.org/10.1128/mBio.00470-18

Lüdecke, C., Jandt, K.D., Siegismund, D., Kujau, M.J., Zang, E., Rettenmayr, M., Bossert, J., Roth, M., 2014. Reproducible biofilm cultivation of chemostat-grown *Escherichia coli* and investigation of bacterial adhesion on biomaterials using a non-constant-depth film fermenter. PLoS One 9. https://doi.org/10.1371/journal.pone.0084837

Maheux, F., Bissonnette, L., Boissinot, M., Bernier, J.T., Huppe, V., Picard, F.J., Be´rube, E., Bergeron, M.G., 2011. Rapid Concentration and Molecular Enrichment Approach for Sensitive Detection of *Escherichia coli* and *Shigella Species* in Potable Water Samples □. Appl. Environ. Microbiol. 77, 6199–6207. https://doi.org/10.1128/AEM.02337-10

Marchesi, J.R., Sato, T., Weightman, A.J., Martin, T.A., Fry, J.C., Hiom, S.J., Wade, W.G., 1998. Design and Evaluation of Useful Bacterium-Specific PCR Primers That Amplify Genes Coding for Bacterial 16S rRNA 64, 795–799.

Meng, J., Zhao, S., Doyle, M.P., 1998. Virulence genes of Shiga toxin-producing *Escherichia coli* isolated from food , animals and humans 45, 229–235.

Molin, S., Tolker-Nielsen, T., 2003. Gene transfer occurs with enhanced efficiency in biofilms and induces enhanced stabilisation of the biofilm structure. Curr. Opin. Biotechnol. 14, 255–261. https://doi.org/10.1016/S0958-1669(03)00036-3

Monte, J., Abreu, A., Borges, A., Simões, L., Simões, M., 2014. Antimicrobial Activity of Selected Phytochemicals against *Escherichia coli* and *Staphylococcus aureus* and Their

Biofilms. Pathogens 3, 473–498. https://doi.org/10.3390/pathogens3020473

Ng, W.-L., Bassler, B.L., 2009. Bacterial Quorum-Sensing Network Architectures. Annu. Rev. Genet. 43, 197–222. https://doi.org/10.1146/annurev-genet-102108-134304

Nordstrom, L., Liu, C.M., Price, L.B., 2013. Foodborne urinary tract infections: A new paradigm for antimicrobial-resistant foodborne illness. Front. Microbiol. 4, 1–6. https://doi.org/10.3389/fmicb.2013.00029

Novick, R.P., Geisinger, E., 2008. Quorum Sensing in Staphylococci. Annu. Rev. Genet. 42, 541–564. https://doi.org/10.1146/annurev.genet.42.110807.091640

Olsen, I., 2015. Biofilm-specific antibiotic tolerance and resistance. Eur. J. Clin. Microbiol. Infect. Dis. 34, 877–886. https://doi.org/10.1007/s10096-015-2323-z

Rutherford, S.T., Bassler, B.L., 2012. Bacterial quorum sensing: Its role in virulence and possibilities for its control. Cold Spring Harb. Perspect. Med. 2, 2–5. https://doi.org/10.1101/cshperspect.a012427

Safadi, R. Al, Abu-Ali, G.S., Sloup, R.E., Rudrik, J.T., Waters, C.M., Eaton, K.A., Manning, S.D., 2012. Correlation between in vivo biofilm formation and virulence gene expression in *Escherichia coli O104:H4*. PLoS One 7. https://doi.org/10.1371/journal.pone.0041628

Schembri, M.A., Kjærgaard, K., Klemm, P., 2003. Global gene expression in *Escherichia coli* biofilms. Mol. Microbiol. 48, 253–267.

Sharma, A., Shivaprasad, D.P., Chauhan, K., Taneja, N.K., 2019. Obliteration of *E. coli* growth and survival in Indian soft cheese (paneer) using multiple hurdles: Phytochemicals, temperature and vacuum. Lwt 114, 108350.

https://doi.org/10.1016/j.lwt.2019.108350

Simmons, K., Islam, M.R., Rempel, H., Block, G., Topp, E., Diarra, M.S., 2016. Antimicrobial Resistance of *Escherichia fergusonii* Isolated from Broiler Chickens 79, 929–938. https://doi.org/10.4315/0362-028X.JFP-15-575

Simmons, K., Rempel, H., Block, G., Forgetta, V., Vaillancourt, R., Malouin, F., Topp, E., 2014. Duplex PCR Methods for the Molecular Detection of *Escherichia fergusonii* Isolates from Broiler Chickens 80, 1941–1948. https://doi.org/10.1128/AEM.04169-13

Štaudová, B., Micenková, L., Bosák, J., Hrazdilová, K., Slaninková, E., Vrba, M., Ševčíková, A., Kohoutová, D., Woznicová, V., Bureš, J., Šmajs, D., 2015. Determinants encoding fimbriae type 1 in fecal *Escherichia coli* are associated with increased frequency of bacteriocinogeny. BMC Microbiol. 15, 1–9. https://doi.org/10.1186/s12866-015-0530-5

Stepanovic, S., Vukovic, D., Dakic, I., Savic, B., Svabic-Vlahovic, M., 2000. A modified microtiter-plate test for quantification of *Staphylococcal* biofilm formation. J. Microbiol. Methods 40, 175–179. https://doi.org/10.1016/s0167-7012(00)00122-6

Stromberg, Z.R., Johnson, J.R., Fairbrother, J.M., et al., 2017. Evaluation of *Escherichia coli* isolates from healthy chickens to determine their potential risk to poultry and human health. *PLoS One*.12, e0180599. Published 2017 Jul 3. doi:10.1371/journal.pone.0180599

Sudheendra, K.R., Basavaraj, P. V, 2018. Analysis of Antibiotic Sensitivity Profile of Biofilm Forming Uropathogenic *Escherichia coli*. J. Nat. Sci. Biol. Med. 9, 175–179. https://doi.org/10.4103/jnsbm.JNSBM

Sumathi, B.R., Gomes, A.R., Krishnappa, G., 2008. Antibiogram profile based dendrogram analysis of *Escherichia coli* serotypes isolated from bovine mastitis. Vet. World 1, 37–39.

Taneja, N.K., Dhingra, S., Mittal, A., Naresh, M., Tyagi, J.S., 2010. *Mycobacterium Tuberculosis* Transcriptional Adaptation, Growth Arrest and Dormancy Phenotype Development is Triggered by Vitamin C. PLoS One 5, 18–24. https://doi.org/10.1371/journal.pone.0010860

Tezel, B.U., Şanlıbaba, and P., 2018. A Major Concern in Food Industry , Contamination Reservoir : Bacterial Biofilm.

Thakur, P., Chawla, R., Tanwar, A., Chakotiya, A.S., Narula, A., Goel, R., Arora, R., Sharma, R.K., 2016. Attenuation of adhesion, quorum sensing and biofilm mediated virulence of carbapenem resistant *Escherichia coli* by selected natural plant products. Microb. Pathog. 92, 76–85. https://doi.org/10.1016/j.micpath.2016.01.001

Touzain, F., Le Devendec, L., de Boisséson, C., Baron, S., Jouy, E., Perrin-Guyomard, A., Blanchard, Y., Kempf, I., 2018. Characterization of plasmids harboring blaCTX-M and blaCMY genes in *E. coli* from French broilers. PLoS One 13, 1–14. https://doi.org/10.1371/journal.pone.0188768

Uhlich, G.A., Gunther IV, N.W., Bayles, D.O., Mosier, D.A., 2009. The CsgA and lpp proteins of an *Escherichia coli O157:H7* strain affect HEp-2 cell invasion, motility, and biofilm formation. Infect. Immun. 77, 1543–1552. https://doi.org/10.1128/IAI.00949-08

Wang, Q., He, Z., Hu, Y., Jiang, Y., Ma, R., Tang, Z., Liang, J., Liu, Z., Huang, Z., 2012. luxS mutant regulation: quorum sensing impairment or methylation disorder? Sensors (Basel). 12, 6155–6175. https://doi.org/10.3390/s120506155

WHO, 2018. WHO | WHO estimates of the global burden of foodborne diseases. Foodborne Dis. Burd. Epidemiol. Ref. Gr. 268.

Xu, A., Scullen, O.J, Sheen, S., Johnson, J.R., Sommers, C.H., 2019. Inactivation of extraintestinal pathogenic *E. coli* clinical and food isolates suspended in ground chicken meat by gamma radiation. *Food Microbiol*. 84, 103264. doi:10.1016/j.fm.2019.103264

Zhu, J., Knottenbelt, S., Kirk, M.L., Pei, D., 2006. Catalytic mechanism of S-ribosylhomocysteinase: Ionization state of active-site residues. Biochemistry 45, 12195–12203. https://doi.org/10.1021/bi061434v

Legends to figure:

Figure 1. (a) Biofilm forming potential of all *E. coli* isolates as determined by crystal violet assay **(b):** Kinetics of biofilm formation in *E. coli* EMC17 along with Biofilm Forming Rate (BFR).

Figure 2. (a) Visualization of biofilm formation for *E. coli* EMC17 using microscopic techniques on glass and stainless steel surfaces at different time points (24-96h). Visual image of glass slide (A–D), Fluorescence imaging on glass slide (E–H), SEM images (I–L) and fluorescence imaging on stainless steel surface (M–P).

(b) Different stages of biofilm formation for *E. coli* EMC17 grown in nutrient broth. Artwork adapted from Rasamiravaka et al., 2015.

Figure 3. (a) Quantification of Quorum Sensing activity of *E. coli* EMC17 determined as a function of AHLs and its comparison with *E. coli* ATCC25922 and *E. coli* EMC 8.

(b) Quantification of EPS produced by *E. coli* EMC17 and relative comparison with *E. coli* ATCC25922 and *E. coli* EMC 8. Significant difference is calculated using one-way ANOVA. p-value ***<0.001, **0.01, *<0.05.

Figure 4: Fold change in relative gene expression of *E. coli* EMC17 biofilm with respect to planktonic cells when grown in nutrient broth for 24h time period. Mean ± SD from 3 replicate is plotted.

Figure 5: In vitro adhesion and invasion assay of *E. coli* EMC17 strain along with *E. coli* MTCC 729 and *E. coli* DH5α in HepG2 cells. Numbers of adherent and internalized bacteria are presented as log 10 of the mean±
standard deviation (SD) CFU/mL of three independent repeats is plotted.

Table 1: Antibiotic sensitivity pattern *E. coli* EMC17

S.No.	Name of Antibiotic	Zone of Inhibition (mm)	Inference
1	Amikacin	28	Sensitive
2	Gentamicin	29	Sensitive
3	Neomycin	26	Sensitive
4	Imipenem	32	Sensitive
5	Clindamycin	0	Resistant
6	Erythromycin	0	Resistant
7	Amoxicillin	0	Resistant
8	Ampicillin	0	Resistant
9	Oxacillin	0	Resistant
10	Piperacillin	19	Intermediate
11	Ciprofloxacin	39	Sensitive
12	Levofloxacin	38	Sensitive
13	Nalidixic Acid	25	Sensitive
14	Sparfloxacin	35	Sensitive
15	Cephalothin	9	Resistant
16	Co-Trimoxazole	30	Sensitive
17	Sulphafurazole	22	Sensitive
18	Tetracycline	24	Sensitive

Highlights:

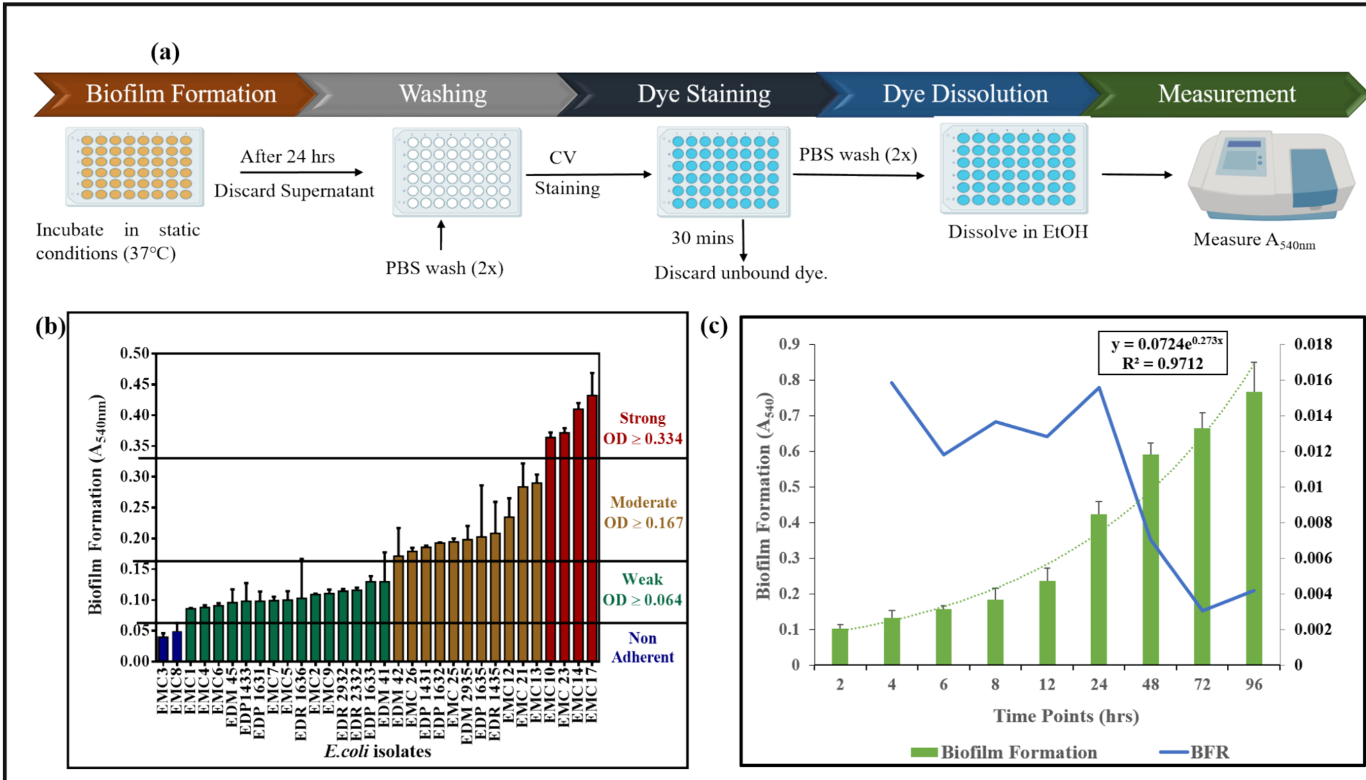
1. Out of 60 products tested 40-60% dairy and 60% meat showed presence of *E. coli*.

2. β-lactam drug resistance was the most commonly observed phenotype.

3. *E.coli* EMC 17 was characterized as MDR, biofilm-forming pathogenic ExPEC strain.

4. EMC17 carries transferrable ESBL gene blaCMY2 escalating AMR spread to other bacteria.

5. Static EMC17 augmented biofilm gene expression, adherence, AHL & EPS production.



(a)
Biofilm Formation
Washing
Dye Staining
Dye Dissolution
Measurement
Incubate in static conditions (37°C)
After 24 hrs
Discard Supernatant
PBS wash (2x)
CV
Staining
30 mins
Discard unbound dye.
PBS wash (2x)
Dissolve in EtOH
Measure A540nm
(b)
Biofilm Formation (A540nm)
Strong
OD ≥ 0.334
Moderate
OD ≥ 0.167
Weak
OD ≥ 0.064
Non
Adherent
E.coli isolates
(c)
y = 0.0724e0.273x
R² = 0.9712
Biofilm Formation (A540)
Time Points (hrs)
Biofilm Formation
BFR

Figure 1

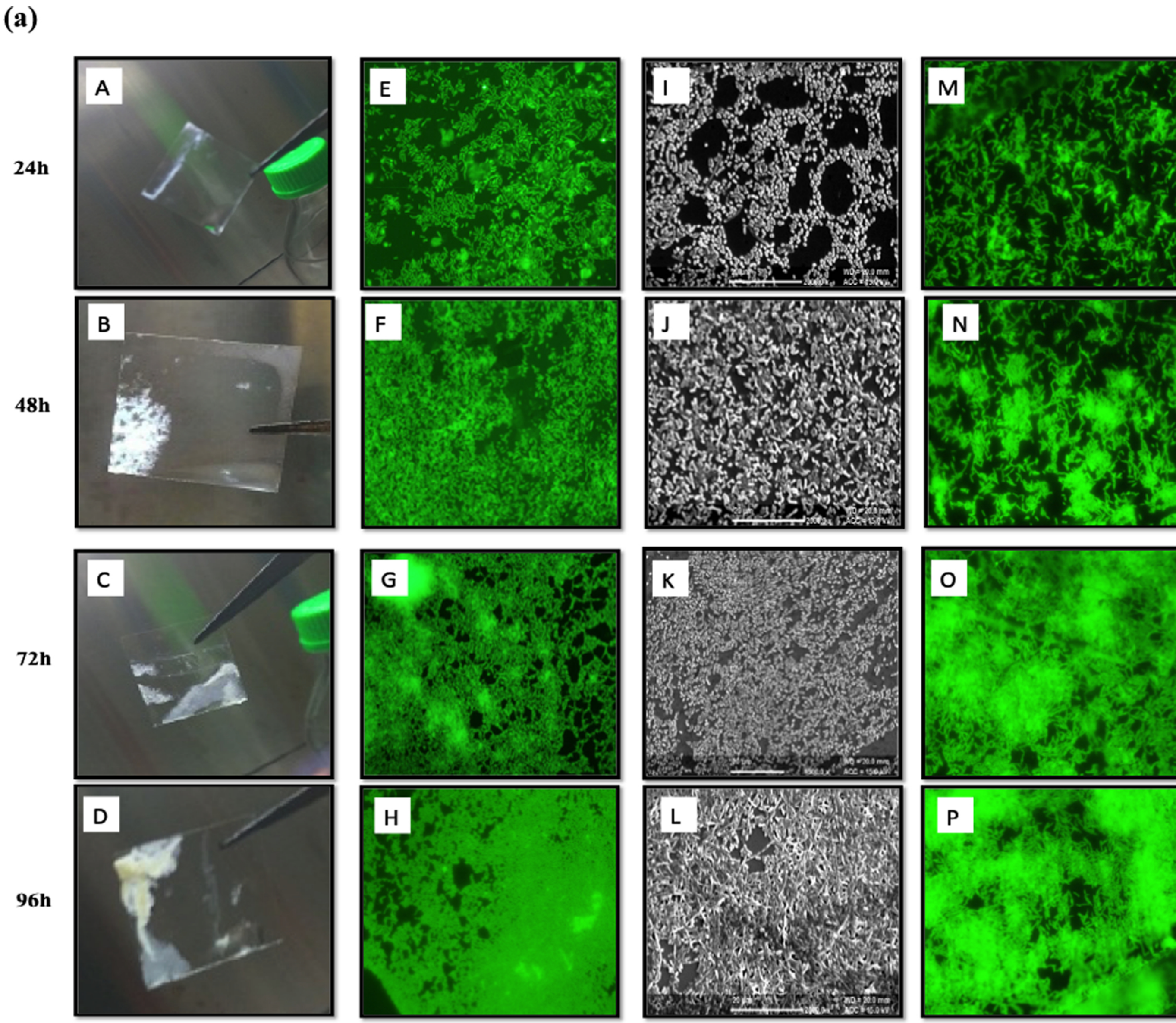
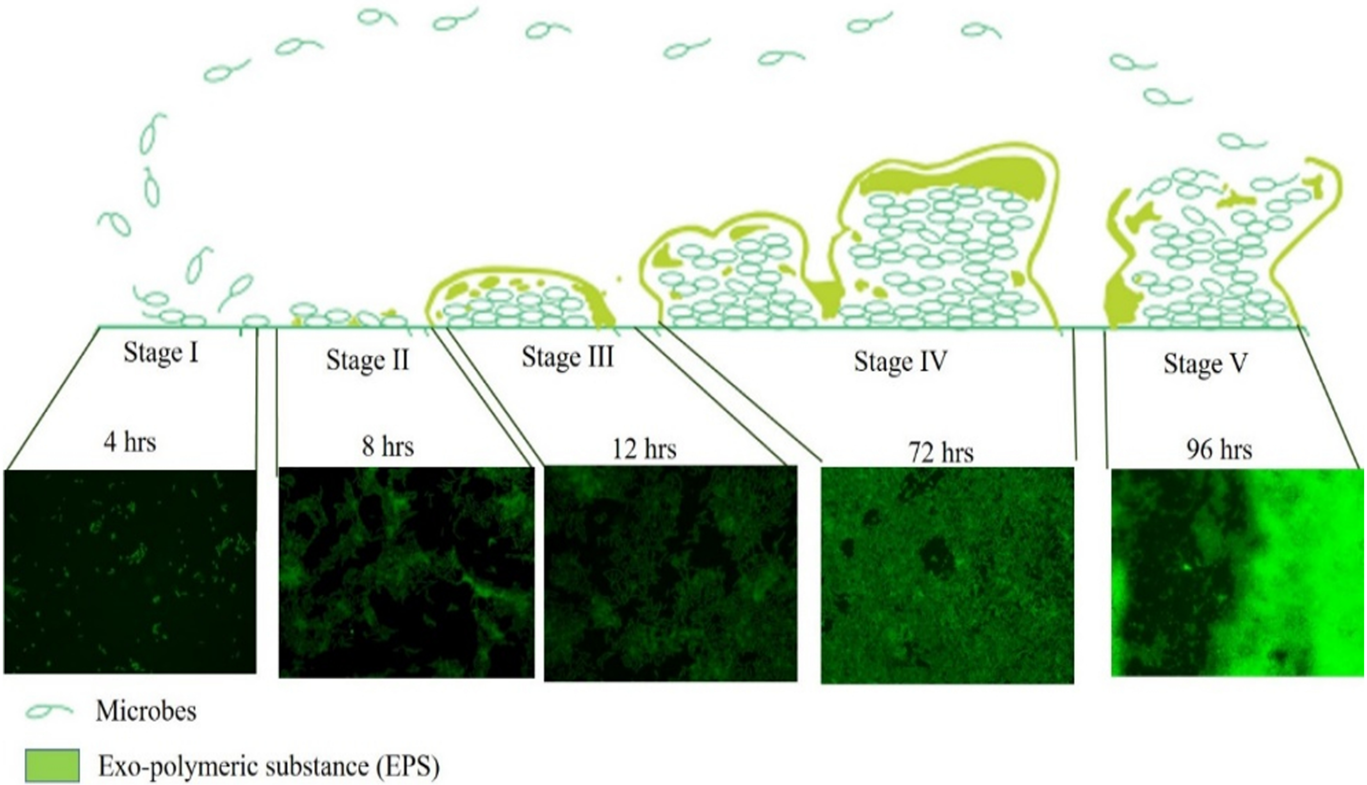


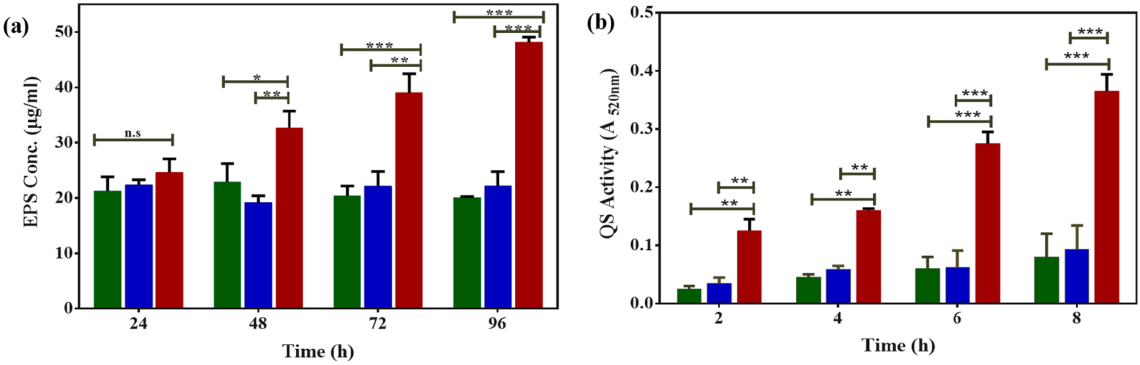
Figure 2a

(b)



Stage I
Stage II
Stage III
Stage IV
Stage V
4 hrs
8 hrs
12 hrs
72 hrs
96 hrs
Microbes
Exo-polymeric substance (EPS)

Figure 2b



(a)
EPS Conc. (µg/ml)
0
10
20
30
40
50
n.s
*
**
***
**
***
***
24
48
72
96
Time (h)
(b)
QS Activity (A 520nm)
0.0
0.1
0.2
0.3
0.4
0.5
**
**
**
**
***
***
***
***
2
4
6
8
Time (h)

Figure 3

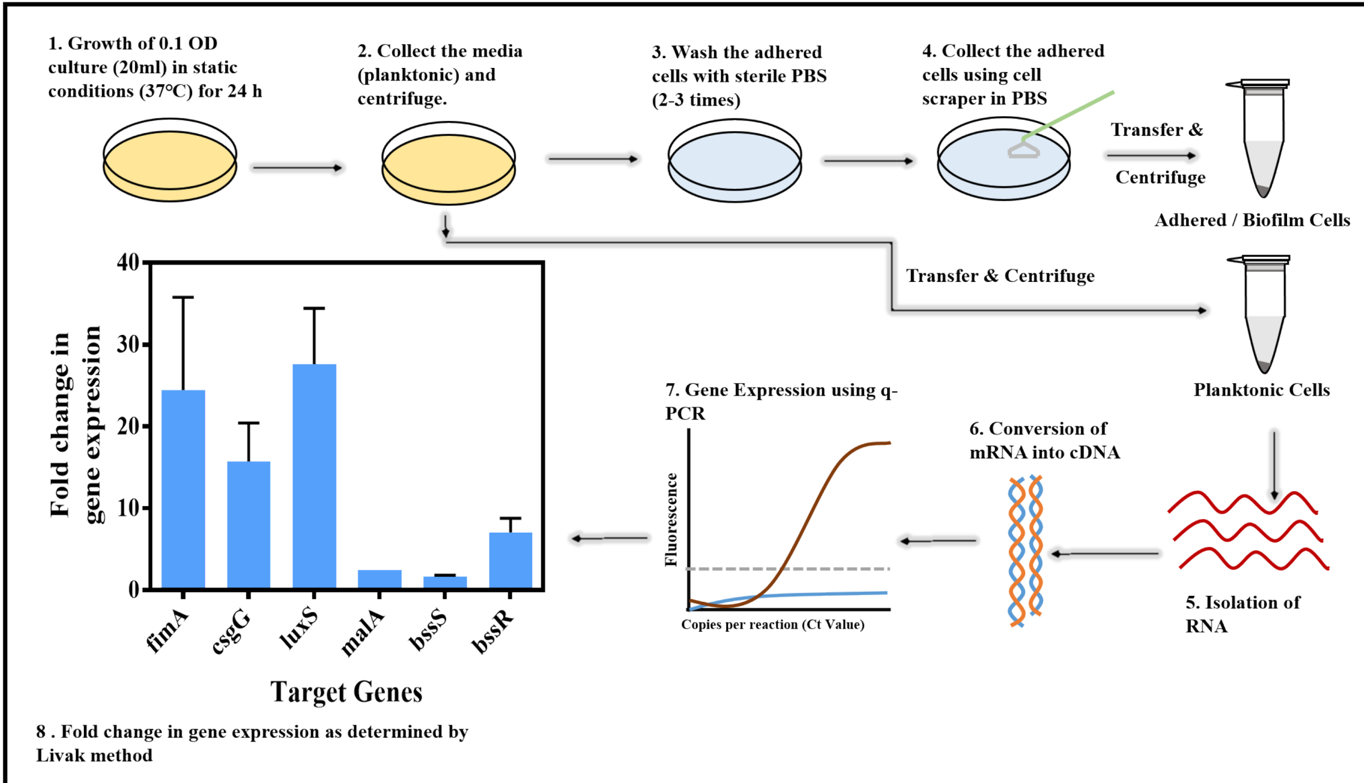
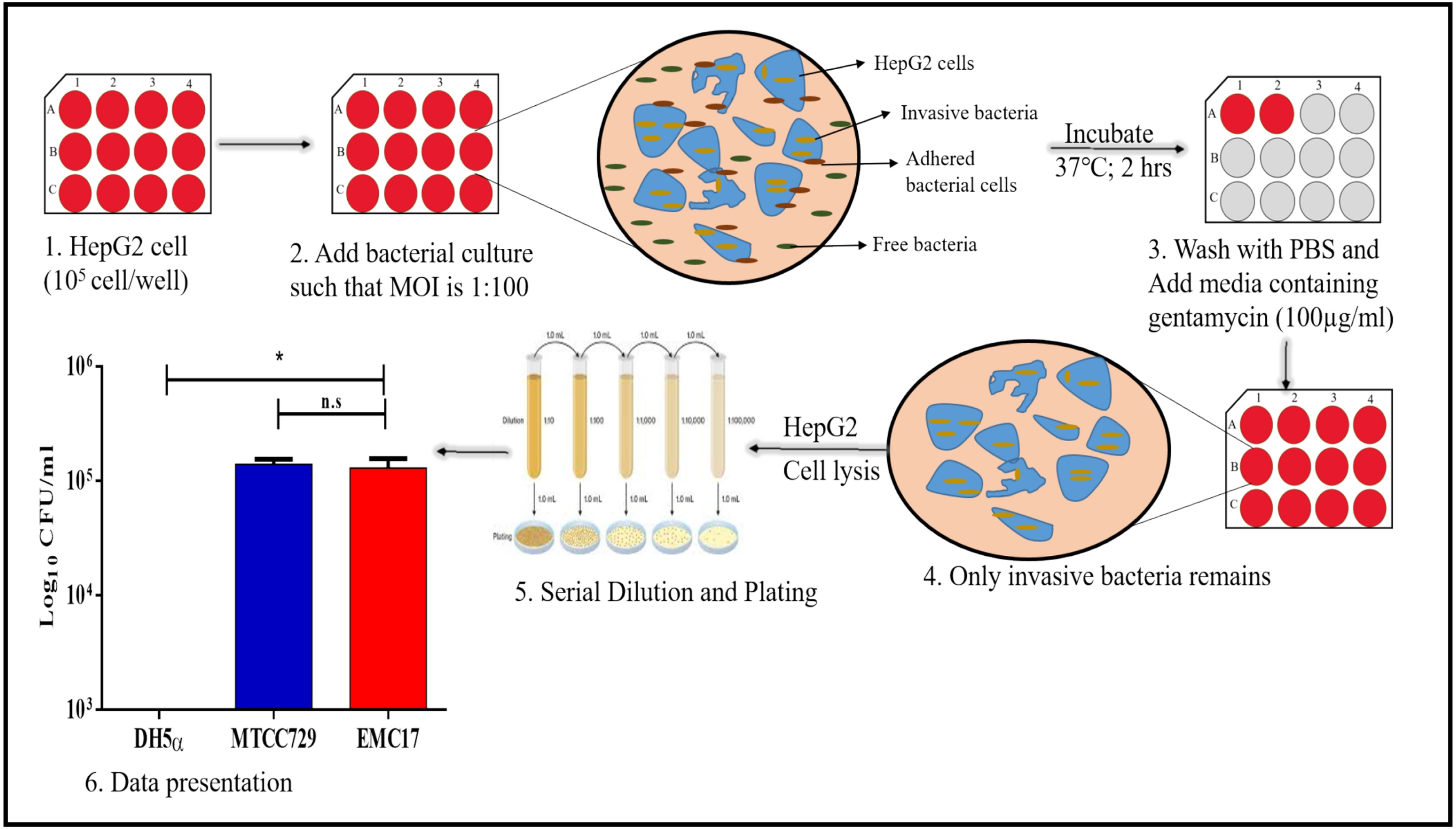

1. Growth of 0.1 OD culture (20ml) in static conditions (37°C) for 24 h
2. Collect the media (planktonic) and centrifuge.
3. Wash the adhered cells with sterile PBS (2-3 times)
4. Collect the adhered cells using cell scraper in PBS
Transfer &
Centrifuge
Adhered / Biofilm Cells
Transfer & Centrifuge
Planktonic Cells
5. Isolation of RNA
6. Conversion of mRNA into cDNA
7. Gene Expression using q-PCR
Fluorescence
Copies per reaction (Ct Value)
Fold change in gene expression
40
30
20
10
0
fimA
csgG
luxS
malA
bssS
bssR
Target Genes
8 . Fold change in gene expression as determined by Livak method

Figure 4



HepG2 cells
Invasive bacteria
Adhered bacterial cells
Free bacteria
1. HepG2 cell (10^5 cell/well)
2. Add bacterial culture such that MOI is 1:100
Incubate
37°C; 2 hrs
3. Wash with PBS and Add media containing gentamycin (100µg/ml)
4. Only invasive bacteria remains
HepG2
Cell lysis
5. Serial Dilution and Plating
Log10 CFU/ml
n.s
DH5α
MTCC729
EMC17
6. Data presentation

Figure 5